# Algebra Mastery Pack — Practice Test A

Grade 11 algebra, functions & data analysis · 24 questions · suggested time 45 minutes · no calculator needed

Name: _____ Date: _____
Score: _____ / 24

**Instructions.** Write full working in the space under each question; answers alone do not show how you got there. Circle one letter for a multiple-choice question and write a single value on the answer line for a short-answer question. Answer every question — a blank scores zero. The answer key with a worked solution for every question is a separate PDF; mark your own paper only after you finish.

**1.** [Solving Linear Equations (for x)]

Solve $2x + 6 = 14$.

A) $x = 3$
B) $x = 4$
C) $x = 8$
D) $x = 10$

**2.** [Literal Equations & Solving for y]

Solve $y = mx + b$ for b.

A) $b = y - mx$
B) $b = y + mx$
C) $b = mx - y$
D) $b = y \div mx$

**3.** [Linear Inequalities]

Solve $x + 3 < 7$.

A) $x < 4$
B) $x < 10$
C) $x > 4$
D) $x \leq 4$

**4.** [Systems of Linear Equations]

Solve the system: $y = 2x$ and $x + y = 9$.

A) (3, 6)
B) (6, 3)
C) (3, 3)
D) (4, 5)

**5.** [Functions: Notation, Domain & Range]

If $f(x) = 3x + 1$, what is $f(2)$?

A) 5
B) 6
C) 7
D) 8

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**6.** [Linear Functions: Slope, Intercepts & Equations]

In the equation $y = 2x + 1$, what is the slope?

A) 1
B) 2
C) $2x$
D) 0

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**7.** [Quadratic Functions & Equations]

Solve $x^2 + 5x + 6 = 0$ by factoring.

A) $x = -2$ or $-3$
B) $x = 2$ or $3$
C) $x = -2$ or $3$
D) $x = 5$ or $6$

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**8.** [Polynomials: Operations & Factoring]

Add: $(3x^2 + 2x + 1) + (x^2 - x + 4)$.

A) $4x^2 + x + 5$
B) $4x^2 + 3x + 5$
C) $3x^2 + x + 5$
D) $4x^2 + x + 3$

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**9.** [Radical & Rational Expressions]

Simplify $\sqrt{50}$.

A) $5\sqrt{2}$
B) 25
C) $10\sqrt{5}$
D) $2\sqrt{25}$

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**10.** [Exponential Functions & Growth/Decay]

In $y = 3 \cdot 2^x$, what is the initial value (the value when $x = 0$)?

A) 3
B) 2
C) 6
D) 0

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**11.** [Data: Measures of Center & Spread]

Find the mean of {2, 4, 4, 6, 9}. Enter a number.

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Answer: _____

**12.** [Scatter Plots, Line of Best Fit & Correlation]

If y tends to rise as x rises, the correlation is ___.

A) positive
B) negative
C) none
D) undefined

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**13.** [Absolute Value Equations & Inequalities]

Solve $|x| = 7$.

A) $x = 7$ only
B) $x = -7$ only
C) $x = 7$ or $x = -7$
D) No solution

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**14.** [Arithmetic & Geometric Sequences]

What is the next term of the arithmetic sequence 4, 7, 10, 13, …?

A) 14
B) 15
C) 16
D) 17

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**15.** [Inverse Functions]

If $f(2) = 9$, what is $f^{-1}(9)$?

A) 2
B) 9
C) 1/9
D) $-9$

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**16.** [Function Transformations & Vertex Form]

The graph of $y = f(x) + 3$ is the graph of $y = f(x)$…

A) shifted up 3 units
B) shifted down 3 units
C) shifted right 3 units
D) stretched vertically by 3

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**17.** [Systems of Linear Inequalities & Linear Programming]

Which point is NOT a solution of the system $y > x - 1$ and $y \leq 2x$?

A) (1, 2)
B) (0, 0)
C) (2, 3)
D) (−1, 0)

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**18.** [Probability from Two-Way Tables]

Using the table, what is P(a student rides the bus)?

A) 0.30
B) 0.40
C) 0.75
D) 0.80

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**19.** [Quadratic Functions & Equations]

Solve $x^2 - 4x - 5 = 0$ by factoring.

A) $x = 5$ or $-1$
B) $x = -5$ or 1
C) $x = 5$ or 1
D) $x = -4$ or 5

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**20.** [Polynomials: Operations & Factoring]

Multiply: $(2x)(3x)$.

A) $6x^2$ B) $5x^2$

C) $6x$ D) $5x$

**21.** [Radical & Rational Expressions]

Solve $\sqrt{x} = 5$. Enter the value of x.

Answer: _____

**22.** [Exponential Functions & Growth/Decay]

For $y = 2^x$, find y when x = 3. Enter a number.

Answer: _____

**23.** [Data: Measures of Center & Spread]

Find the median of {2, 4, 4, 6, 9}. Enter a number.

Answer: _____

**24.** [Scatter Plots, Line of Best Fit & Correlation]

If y tends to fall as x rises, the correlation is ___.

A) negative B) positive

C) none D) undefined